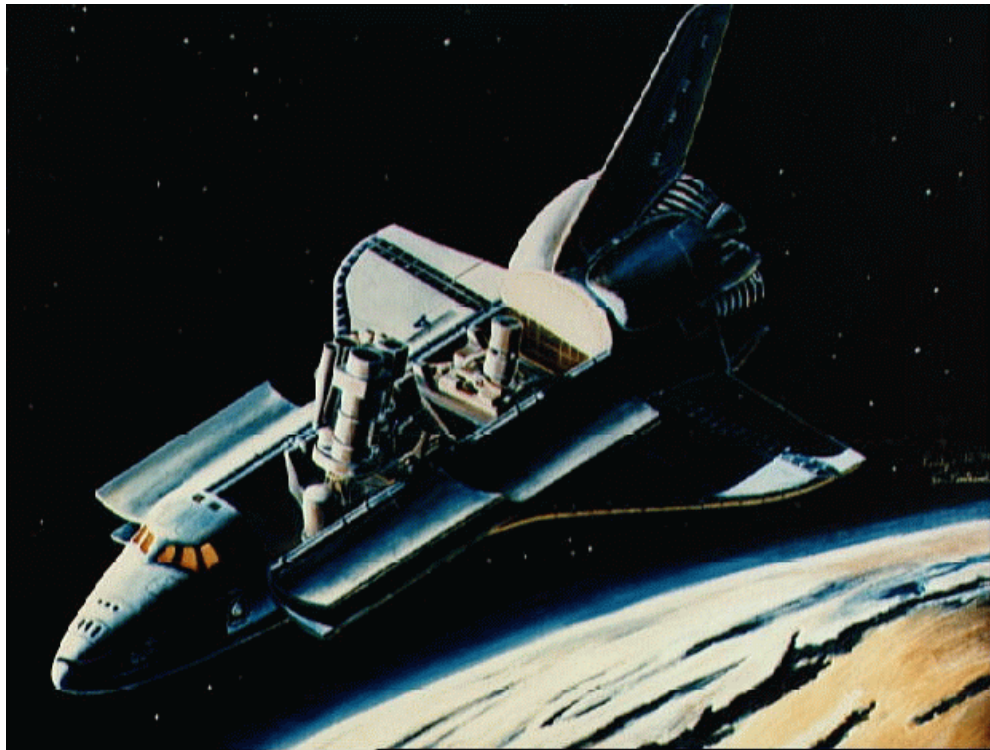


THE ASTRO MISSIONS

ULTRAVIOLET IMAGING TELESCOPE



Astro-1 Mission on orbit (artist's concept)

BACKGROUND

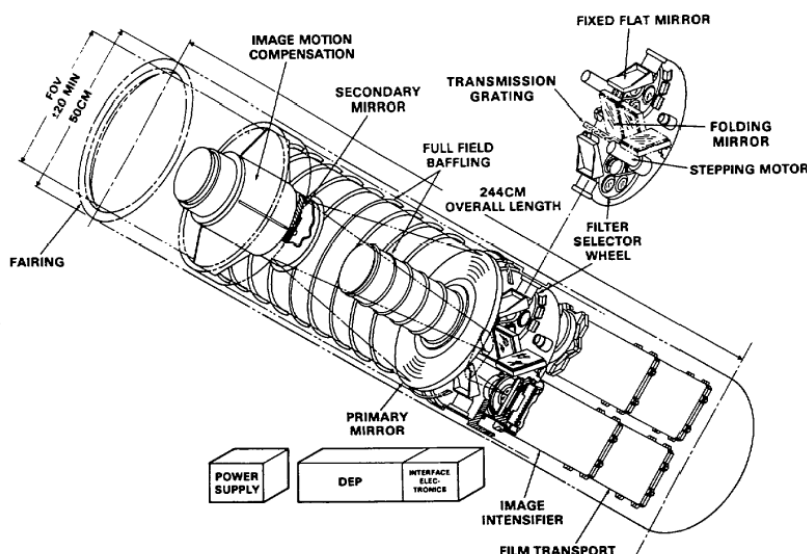
The Ultraviolet Imaging Telescope, one of the three ultraviolet telescopes of the *Astro* Observatory, flew on Space Shuttle missions in December 1990 and March 1995. It was designed, built, and operated by a team led by Theodore P. Stecher at Goddard Space Flight Center.

UIT was proposed in 1978 as part of a program to fly small ("university-class") scientific experiments on the "Spacelab" modules attached to the Space Shuttle. At that time, NASA anticipated that there would be about 3 Shuttle/Spacelab missions each year dedicated to astrophysics experiments (out of an envisioned 20 or so missions annually). By late 1985, the Astro UV telescopes had been approved for six missions. At the time of the Challenger

accident in January 1986, *Astro-1* was in the Shuttle Columbia payload bay ready for the launch to follow the Challenger mission. In the aftermath of the accident and in the face of the rapidly escalating costs of the Shuttle itself, the Spacelab astrophysics program was drastically reduced. Of the 200 experiments proposed in 1978, only the four telescopes of the *Astro-1* mission were actually flown.

The *Astro* Observatory was intended to be a UV pathfinder for the Hubble Space Telescope, but rescheduling caused by the accident meant that HST was launched first.

UIT was a 38-cm Cassegrain telescope carrying two ultraviolet cameras with 40 arc-minute diameter fields of view (150 times larger than the field of the Hubble Space Telescope). The cameras operated in the "far-UV" (120-200 nm) and "mid-UV" (200-320 nm) spectral ranges, respectively, and carried a total of 12 filters. They employed dual-stage, magnetically focussed image intensifiers with very strong long-wavelength rejection to provide pure UV response. Images produced by phosphors at the output of the intensifiers were recorded on film. (At the time UIT was approved, there were no proven electronic detectors capable of covering the required large, 2048x2048 pixel field of view.) Fine guidance was accomplished with an articulated secondary mirror controlled by a signal from an externally mounted 15-cm star tracker containing a CCD detector. Final resolution for point sources was typically 2.7 arc-seconds FWHM. A cutaway drawing of the UIT design is shown below.



During the two *Astro* missions, UIT returned a total of 1570 data frames of 250 different astronomical targets.

The *Astro* UV payload consisted of UIT, the Hopkins Ultraviolet Telescope (HUT), and the Wisconsin Ultraviolet Photo-Polarimeter Experiment (WUPPE). HUT was a far-ultraviolet spectrometer (Johns Hopkins University, led by A. F. Davidsen). WUPPE was a UV spectro-polarimeter (University of Wisconsin, led by A. D. Code). These three instruments were co-aligned and mounted on the Spacelab Instrument Pointing System. They were operated simultaneously on a given target by crew members from the Shuttle aft flight deck.

For *Astro-1* ([STS-35](#), 1990, Shuttle Columbia) the Broad-Band X-Ray Telescope (BBXRT, P. Serlemitsos, Goddard Space Flight Center) was operated independently on a separate

pointing system. *Astro-1* compiled a NASA record for delays: 33 different assigned launch dates over 8 years, 12 complete target timelines planned (each 8 days long at 1 second resolution), and 4 scrubs within 6 hours of launch.

On *Astro-2* ([STS-67](#), 1995, Shuttle Endeavour), the UV telescopes flew alone. *Astro-2* was the first NASA mission to have an active Internet home page and logged over 2.5 million hits from over 200,000 people during its 15 day duration.

A list of over 140 publications based on data from the UIT can be found [here](#).

MISSION IMAGES

[\[Click for Full Size\]](#)



UIT Design



Vacuum-Tank Testing
(HUT)



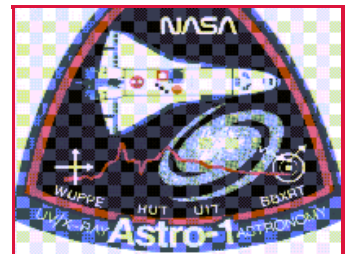
UV Instruments
On Pallet



Co-Mounted UV
Telescopes



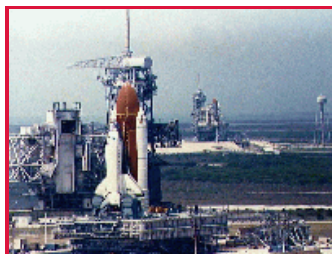
Astro-1 Payload
On Pallets



Astro-1 Patch



Rollback Of Astro
Rollout Of HST



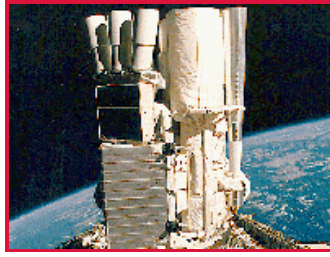
Astro-1 & HST On
Launch Pads



On The Launch Pad



Astro-1 Launch



UIT On Orbit



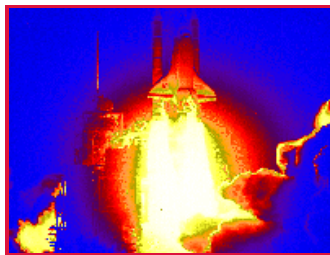
Astro-1 Crew



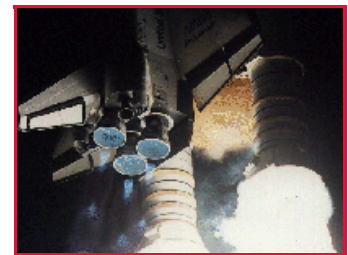
Lunch

Moon Rising Over
Earth's LimbOrion Rising
(Daylight)

Astro-2 Patch



Astro-2 Launch



Main Engines At Launch



Astro-2 Crew

Astro-2 Payload
On Orbit

Payload Into Shadow



Stars over the Limb

Aft Flight Deck
(Instrument Control)

Running the Instruments



Orbiting Target Book



Payload Operations Control
Center (Huntsville, AL)



Over the Andes



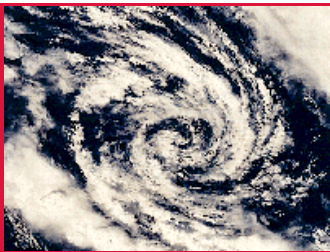
Disappointment Reach
(Australia)



Nepal



Oahu



Low Pressure System



Sunset



The Dreaded Exercycle



Landing Practice



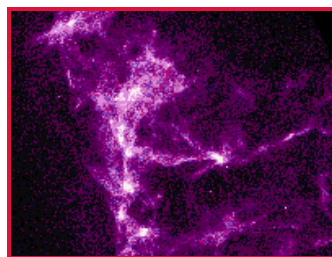
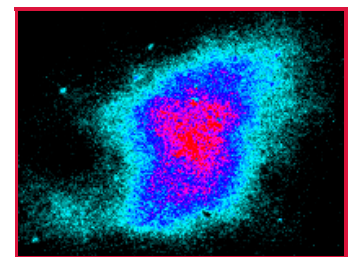
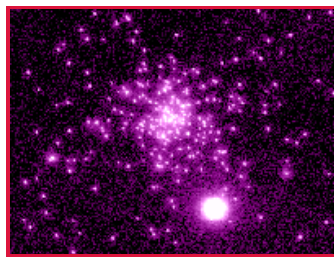
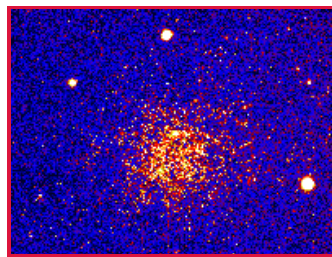
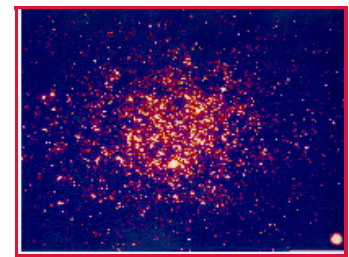
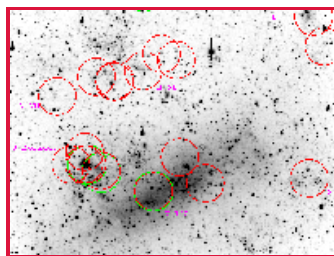
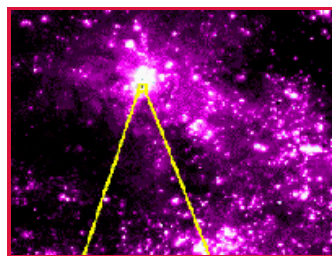
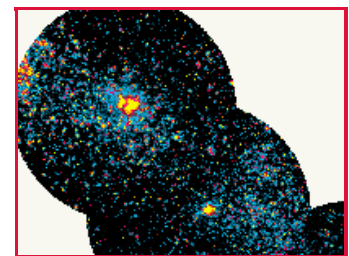
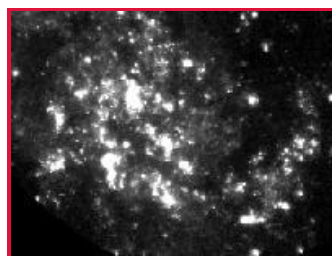
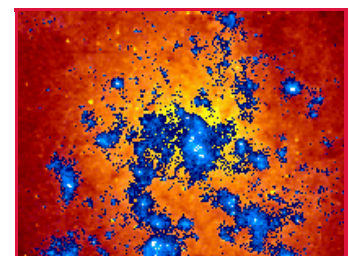
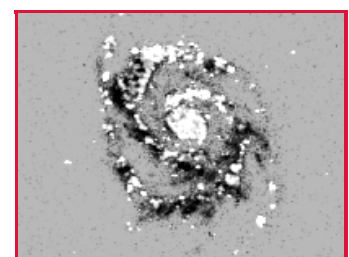
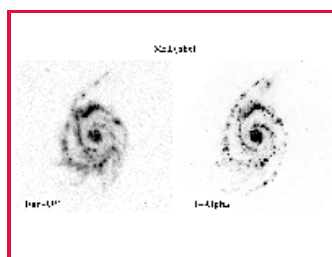
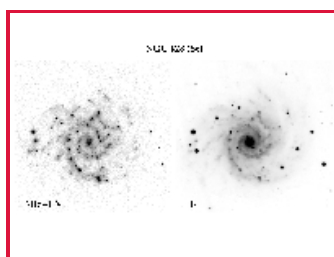
Landing (Edwards
Air Force Base)

SCIENCE IMAGES

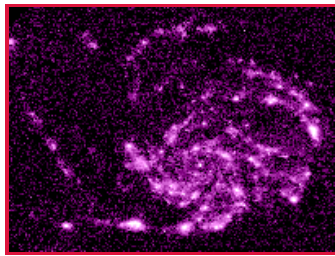
[Far-Ultraviolet Unless Noted;
Click for Full Size]



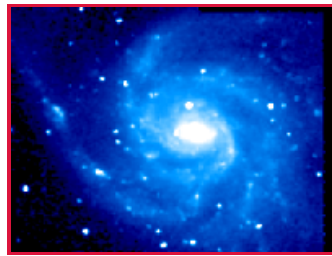
Moon (Far-UV, Optical)

Cygnus Loop
(Supernova Remnant)Crab Nebula 4 Bands
(Supernova Remnant)NGC 6752
(Globular Cluster)Omega Centauri
(Globular Cluster)Omega Centauri
(Center)UIT Fields in the
Large Magellanic Cloud3 Views of
30 Doradus (LMC)Small Magellanic Cloud
MosaicM31 (Andromeda Spiral)
Center and SW Disk
(Mid-UV Mosaic)M33 (Spiral; UV/Optical
Comparison, 4 Bands)M33
UV/Optical Superposition

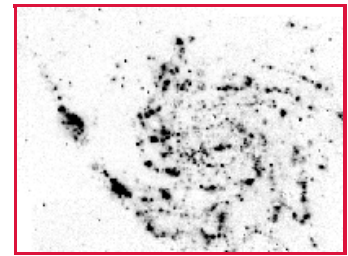
M74 (Spiral;
Mid-UV, Optical)



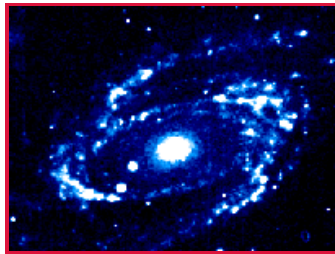
M51 (Spiral;
Far-UV, H-alpha)



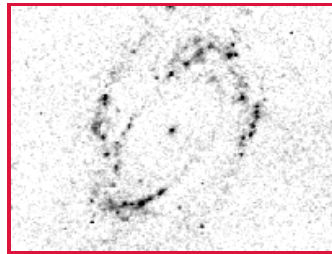
M51 (Far-UV, H-alpha
Color Map)



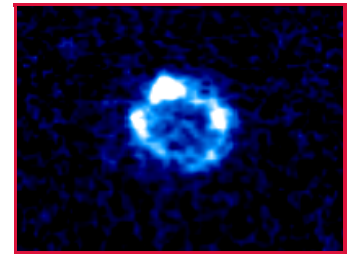
M101 (Spiral
Galaxy)



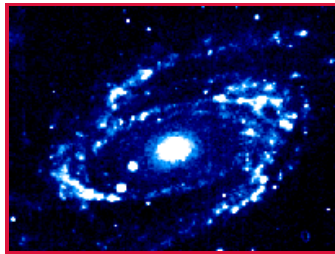
M101 (Far-UV,
Optical)



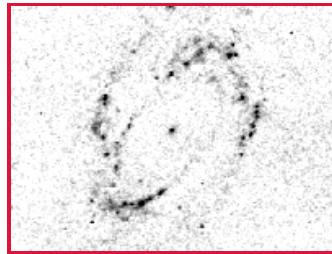
M101 (Far-UV,
H-alpha)



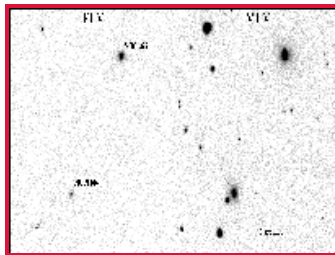
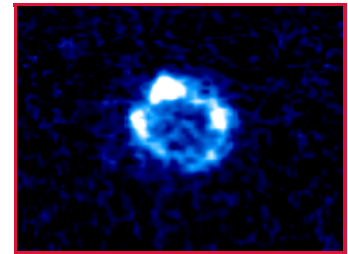
M81 (Spiral;
Mid-UV, Optical)



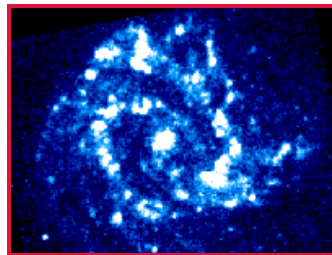
M81 (Far-UV,
Mid-UV, Optical)



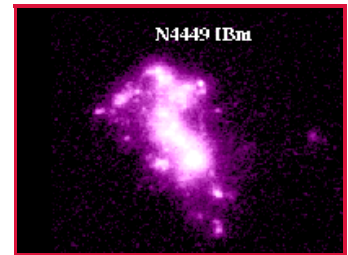
NGC 1317 (Sa Galaxy
With Nuclear Ring)



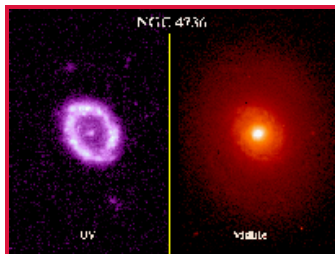
NGC 1399/1404
(E Galaxies)



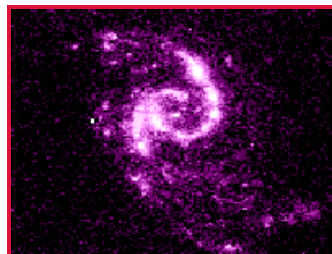
M83 (Barred Spiral;
Mid-UV With
UIT Rocket Prototype)



Irregular & Dwarf
Galaxies



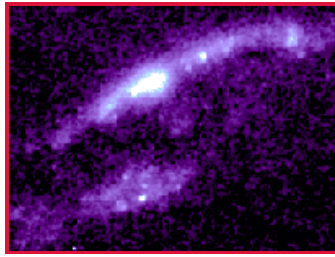
NGC 4736 (Spiral
With Interior Ring)



3 Spiral Galaxies



M82 (Starburst Galaxy
With Galactic Wind)



Cen A: Post-Merger
Starburst Ring

LINKS

- [Astro Missions: Description & Data Archive](#) (STScI Multi-Mission Archive)
- [Astro-2: A Shuttle Mission Made in Heaven](#) (Bill Blair, JHU)
- [Additional Astro-1 Mission \(STS-35\) Information and Images](#) (NASA)
- [Additional Astro-2 Mission \(STS-67\) Information and Images](#) (NASA)
- [Selected Videos from Astro-2](#) (QuickTime)
- [Astronaut-Narrated Video Summary of Astro-1 Mission](#)
- [Astronaut-Narrated Video Summary of Astro-2 Mission](#)
- [Lecture on Ultraviolet Astronomy](#) (UVA ASTR 511, advanced undergraduate/graduate)
- [Technical Description of UIT](#) (Astrophysical Journal)
- [Technical Description of UIT and UIT Data Reduction](#) (Publications of the Astronomical Society of the Pacific)
- [Overview of UIT Science Programs](#) (for Astro-2 Guest Investigator proposals)
- [Publications based on UIT data](#)
- [HUT home page](#)
- [WUPPE home page](#) (U. Wisconsin)
- [GALEX Mission](#) (long-duration, wide field, UV imaging satellite mission, launched 2003)

Last modified **September 2024** by [R. W. O'Connell](#)

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